Business Requirements Document (BRD)

Earthschell Technologies — Corporate Website

Version: 1.0
Status: Draft — Pending Business Validation
Project Owner: Earthschell Technologies
Project Coordinator: Hanna Imbichi
Project Type: Corporate Website
Reference Website: https://stamod.us/

---

1. Project Overview

Earthschell Technologies aims to establish a professional digital presence that positions the company as a partner for product development, custom engineering solutions, and engineering resource coordination.

The company will serve clients in India and international markets, including Germany and other regions, by understanding their requirements and connecting them with suitable engineering expertise, service providers, companies, and manufacturing resources.

The website will communicate Earthschell's capabilities, explain its service offerings, establish credibility, and help potential clients initiate project discussions.

The design will take inspiration from the clarity, positioning, and professional presentation of STAMOD while maintaining an original brand identity and website experience.

2. Business Objectives

- Establish a credible international presence for Earthschell Technologies.
- Clearly communicate the company's engineering and product development services.
- Attract business enquiries from international clients and individual innovators.
- Connect clients with suitable engineering capabilities and manufacturing resources.
- Showcase the company's expertise, approach, and approved project experience.
- Generate qualified enquiries through a clear project enquiry process.
- Create a scalable digital platform that can accommodate additional services as the business grows.

3. Business Model

Earthschell Technologies will operate as an engineering solutions partner, helping clients move from a product requirement or engineering challenge towards an appropriate solution.

The business may coordinate with external engineering companies, technical specialists, machinery providers, and manufacturing partners based on the requirements of each project.

The website must explain the value Earthschell provides in this process, such as requirement understanding, solution coordination, partner identification, and project communication, where these activities are confirmed as part of its actual operating model.

The company must clarify whether it acts as a direct engineering service provider, a project coordinator, a manufacturing sourcing partner, or a combination of these roles.

4. Target Audience

4.1 International Business Clients

Companies seeking engineering support, product development, technical expertise, manufacturing capabilities, or suitable project partners.

4.2 Automotive and Industrial Companies

Organisations requiring product design, engineering support, custom technical solutions, or manufacturing coordination.

4.3 Startups and Product Innovators

Individuals and early-stage businesses seeking support to develop a product concept or identify suitable engineering and manufacturing resources.

4.4 Engineering and Manufacturing Partners

Companies and service providers that may collaborate with Earthschell to fulfil client requirements.

5. Core Service Categories

The following are proposed service categories and must be validated with the business owner before publication.

5.1 Product Development

Support for clients developing new products or improving existing product concepts.

Potential capabilities, subject to confirmation:

- Product concept development.
- Product design and engineering.
- CAD modelling and technical design.
- Prototyping coordination.
- Design validation and development support.

5.2 Custom Engineering Solutions

Tailored engineering support based on specific client requirements.

Potential capabilities:

- Custom engineering project coordination.
- Technical requirement analysis.
- Identification of suitable engineering resources.
- Coordination with specialised engineering partners.
- Support for project-specific technical challenges.

5.3 Engineering and Manufacturing Network

Connecting client requirements with suitable companies, specialists, equipment, and manufacturing resources.

Potential capabilities:

- Engineering partner identification.
- Manufacturing partner coordination.
- Machinery and technical resource sourcing.
- Supplier or capability discovery.
- Communication and coordination between stakeholders.

5.4 Additional Services

The website should support future service categories as Earthschell expands into additional engineering, industrial, product development, or manufacturing-related opportunities.

Only services that Earthschell can genuinely deliver or coordinate should be published.

6. Project Scope

In Scope

- Corporate website design and development.

- Responsive layouts for desktop, tablet, and mobile.
- Homepage and company introduction.
- Product development service page.
- Custom engineering solutions service page.
- Engineering and manufacturing network information.
- About the company.
- Projects or case studies, where approved material is available.
- Contact and project enquiry form.
- Clear calls to action for prospective clients.
- Basic SEO and performance optimisation.
- Privacy notice and terms of use.
- Website testing and production deployment.

Out of Scope for Initial Release

- Client accounts and login.
- Online payments.
- Automated quotation generation.
- Client project management dashboards.
- Supplier portals or partner registration systems.
- Machinery inventory or live availability tracking.
- Automated supplier matching.
- AI-powered engineering tools.
- Learning management systems or training libraries.

These features may be considered in future phases if supported by a validated business requirement.

7. Website Sitemap

1. **Home**
    - Business value proposition.
    - Services overview.
    - How Earthschell works.
    - Industries and capabilities.
    - Featured projects.
    - Final call to action.
2. **Services**
    - Product Development.
    - Custom Engineering Solutions.
    - Engineering and Manufacturing Coordination.
    - Additional confirmed services.
3. **Industries**
    - Automotive.
    - Manufacturing and industrial sectors.
    - Other confirmed industries.
4. **Projects**

    - Approved case studies.
    - Project challenges, approaches, and outcomes.
5. **About Us**
    - Company profile.
    - Engineering expertise.
    - Business approach.
    - Leadership or team information, where approved.
6. **Contact / Discuss a Project**
    - Project enquiry form.
    - Business contact information.
    - Email and other approved communication channels.
7. **Legal Pages**
    - Privacy Policy.
    - Terms and Conditions.

8. Homepage Requirements

8.1 Hero Section

Communicate the core value proposition immediately, with a primary call to action that encourages prospective clients to discuss a project.

Proposed headline:
“Engineering Expertise. Connected to Your Needs.”

Supporting copy:
“Earthschell Technologies helps businesses and innovators navigate product development, custom engineering, and manufacturing opportunities by connecting project requirements with the right expertise and resources.”

Primary CTA: Discuss Your Project
Secondary CTA: Explore Our Services

8.2 Services Overview

Present the main service categories using concise descriptions and clear navigation to their respective pages.

8.3 How We Work

Explain the proposed engagement process:

1. Understand the client's requirement.
2. Identify appropriate capabilities and resources.
3. Coordinate the proposed solution and stakeholders.
4. Support project execution and communication, as agreed for each engagement.

8.4 Industries and Capabilities

Introduce the sectors and engineering capabilities Earthschell can support, using verified information.

8.5 Projects and Experience

Showcase approved project examples, relevant technical visuals, and measurable outcomes where available.

8.6 Closing CTA

Encourage prospective clients to share their requirements and initiate a discussion.

9. Functional Requirements

FR-01 — Website Navigation: Users must be able to access all primary pages through a consistent navigation structure.

FR-02 — Service Discovery: Users must be able to understand the company's core services and explore detailed descriptions.

FR-03 — Project Enquiry: Users must be able to submit their project requirements through a contact form.

FR-04 — Enquiry Delivery: Form submissions must reach the designated company email or approved enquiry management system.

FR-05 — Form Validation: Required fields, email formats, errors, and successful submissions must be handled appropriately.

FR-06 — Responsive Design: The website must adapt to common mobile, tablet, and desktop screen sizes.

FR-07 — International Accessibility: Content must use clear professional English and support enquiries from international clients.

FR-08 — Portfolio Presentation: Approved project information must be presented in a clear and structured format.

FR-09 — SEO: Pages must support descriptive titles, meta descriptions, appropriate URLs, and search engine indexing.

FR-10 — Legal Information: Privacy and terms pages must be accessible from the website footer.

10. Non-Functional Requirements

Usability

- Clear information hierarchy.
- Concise, professional content.
- Consistent navigation and calls to action.
- Simple access to services and contact information.

Performance

- Optimised images and assets.
- Efficient loading across typical network conditions.
- Limited unnecessary animation and third-party scripts.

Accessibility

- Readable typography and adequate contrast.
- Semantic headings and form labels.
- Alternative text for meaningful images.
- Keyboard-accessible navigation and controls.

Security and Privacy

- HTTPS for the live website.
- Secure handling of enquiry submissions.
- Appropriate spam protection.
- Responsible collection and processing of personal information.
- Accurate privacy and terms documentation.

Scalability

- A structure that allows additional services, industries, and case studies to be added without redesigning the entire website.

11. Brand and Design Direction

The visual identity should communicate technical expertise, precision, reliability, and international professionalism.

Design principles:

- Light backgrounds balanced with dark charcoal sections.
- Neutral industrial colours with a restrained accent colour.
- Modern, readable typography.
- High-quality automotive, product engineering, and manufacturing imagery.

- Clean layouts with generous whitespace.
- Subtle transitions and purposeful motion.
- Consistent components and responsive layouts.

The STAMOD reference should inform the overall clarity and engineering-oriented presentation, not be copied directly.

12. Assumptions and Dependencies

- The company will confirm its exact service offerings and operating model.
- The business owner will provide approved company information and contact details.
- Engineering capabilities and partner arrangements will be validated before publication.
- Project visuals and case studies will be approved for public use.
- The company will confirm the intended international markets and preferred enquiry process.
- The domain, hosting, and enquiry delivery mechanism will be configured before launch.
- Legal documents will be reviewed for the company's actual business and data practices.

13. Success Metrics

The website's effectiveness may be assessed through:

- Number of relevant project enquiries received.
- Number of qualified business enquiries.
- Enquiry conversion rate.
- Traffic to service pages.
- Visits from intended international markets.
- Website usability and performance.
- Successful completion of the agreed acceptance criteria.

Targets should be established after baseline traffic and business expectations are understood.

14. Acceptance Criteria

The initial release will be accepted when:

1. All approved pages are implemented.
2. Content accurately represents the company's services and operating model.
3. The website follows the approved visual identity.
4. Responsive behaviour is verified across common screen sizes.
5. Enquiry forms validate correctly and deliver submissions.
6. Navigation and external links function correctly.

7. SEO metadata and basic accessibility checks are completed.
8. Legal pages are published following appropriate review.
9. No unresolved critical defects remain.
10. The business owner approves the final website.

15. Open Questions and Decisions

- Does Earthschell deliver engineering work directly, coordinate external partners, or do both?
- Which product development and engineering capabilities are confirmed?
- Does the company have an existing network of manufacturing companies and machinery providers?
- Which countries and industries should be prioritised initially?
- Will Earthschell manage project delivery or primarily connect clients with providers?
- Are there approved projects or case studies to publish?
- Who will receive and manage incoming enquiries?
- Is a content management system required?
- Who approves website content and design?
- What is the target launch date?

16. Approval

Business Owner: Earthschell Technologies
Project Coordinator: Hanna Imbichi
Document Status: Draft — Pending Validation

This BRD defines the proposed requirements for the initial corporate website. Final scope and published claims must be approved by the business owner before development and launch.